

Airbnb Pricing Intelligence & Revenue Optimization Analysis

Executive Summary:

Property management companies often face challenges in determining competitive listing prices and identifying markets with the greatest revenue potential. This project analyzes the New York City Airbnb dataset to provide data-driven insights into pricing dynamics, customer demand, neighborhood performance, and market opportunities that can support strategic portfolio and pricing decisions. The analysis was conducted using the NYC Airbnb Open Data dataset, containing information on listing prices, room types, neighbourhoods, host characteristics, availability, and customer reviews. Data preparation included handling missing values, removing extreme price outliers using the 99th percentile, and creating additional features to support business analysis. Customer demand was represented using reviews per month, as the dataset did not contain booking or occupancy information. The analysis revealed that Manhattan remains New York City's strongest premium market, commanding the highest listing prices while maintaining healthy customer demand. A deeper neighborhood-level analysis identified areas such as SoHo, Chelsea, Financial District, and Theater District as attractive markets due to their combination of pricing power, customer engagement, and market size. In contrast, Tribeca demonstrated exceptionally high prices but lower demand, indicating a niche luxury market. Room type analysis further showed that private rooms generate the highest customer engagement, while entire homes command the highest prices. Statistical testing confirmed that customer engagement differs significantly across room types. To support pricing decisions, three predictive models were evaluated: Linear Regression, Random Forest, and XGBoost. XGBoost achieved the strongest predictive performance (Cross-Validation $R^2 \approx 0.63$, Test $R^2 \approx 0.65$), demonstrating its ability to capture the complex, non-linear relationships influencing Airbnb pricing.

2. Business Objective

Background:

The New York City Airbnb market is highly competitive, with listing prices and customer demand varying significantly across neighbourhoods and property types. Property management companies need reliable market insights to determine where to expand their portfolio, how to price listings competitively, and which markets offer the greatest revenue potential. Making these decisions without data can result in missed opportunities or ineffective pricing strategies.

Problem Statement:

Property managers need to understand the key factors influencing Airbnb listing prices and customer demand to make informed investment and pricing decisions. However, variations in location, room type, host characteristics, and market competition make these decisions

challenging. This project aims to identify the factors driving pricing, evaluate demand patterns across New York City, and uncover neighbourhoods with the greatest business potential.

Stakeholder:

The primary stakeholder for this project is the Property Management Team, responsible for acquiring new properties, optimizing pricing strategies, and identifying high-potential markets for portfolio expansion.

Objectives:

- The objectives of this analysis are to:
 - Analyze how the New York City Airbnb market is segmented by listing price.
 - Identify neighbourhood groups and neighbourhoods that command premium pricing.
 - Evaluate customer demand using reviews per month as a demand proxy.
 - Understand how room type influences pricing and customer engagement.
 - Identify the factors that have the greatest influence on listing prices.
 - Develop a predictive model to estimate Airbnb listing prices.
 - Provide data-driven recommendations to support pricing, investment, and portfolio expansion decisions.
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3. Dataset Overview

Data Source

The analysis was conducted using the New York City Airbnb Open Data dataset, publicly available on Kaggle. The dataset contains detailed information on Airbnb listings across the five boroughs of New York City and is widely used for market and pricing analysis.

Dataset Summary:

Attributes	Description
Data Set	NYC Airbnb Open Data
Source	Kaggle
Number of Listings	48,895
Number of Features	16
Target Variable	Price

Key Features

The analysis primarily focused on the following variables:

- Price – Listing price (target variable).

- **Neighbourhood Group** – Borough where the listing is located.
- **Neighbourhood** – Specific area within each borough.
- **Room Type** – Entire home/apartment, private room, or shared room.
- **Minimum Nights** – Minimum number of nights required per booking.
- **Number of Reviews** – Total customer reviews received.
- **Reviews per Month** – Used as a proxy for customer demand.
- **Calculated Host Listings Count** – Number of listings managed by each host.
- **Availability (365)** – Number of days the property is available during the year.

Missing Values

A data quality assessment identified missing values primarily in the `last_review` and `reviews_per_month` columns. Since `last_review` was not required for this business analysis, it was excluded. Missing values in `reviews_per_month` were replaced with 0, assuming that listings without recorded reviews had no recent review activity.

4. Data Cleaning & Preparation

Removed missing values:

A data quality assessment identified missing values primarily in the `last_review` and `reviews_per_month` columns. Missing values (NaN) in `reviews_per_month` were replaced with 0, assuming that listings without recorded reviews had no recent review activity.

Used 99th percentile for outliers:

The price variable contained a small number of extremely high-priced listings, with some values reaching nearly \$10,000. These extreme values could disproportionately influence summary statistics and visualizations, making it difficult to identify general market patterns.

To reduce their impact, the 99th percentile (quantile) method was applied. Listings with prices above the 99th percentile (approximately \$800) were excluded from the analysis. This means that 99% of the listings were retained, while only the most extreme 1% of observations were removed.

This approach preserved the vast majority of the dataset while reducing the influence of extreme luxury listings, resulting in a more representative analysis of the New York City Airbnb market.

Created price categories:

To better understand the Airbnb market, listing prices were grouped into five price categories: Budget, Standard, Premium, High, and Luxury. Instead of analyzing thousands of individual price values, these categories provide a clearer view of how listings are distributed across different market segments. Segmenting prices makes it easier to compare supply, demand, and pricing patterns across neighbourhoods, room types, and host categories. It also helps identify which price segments dominate the market and where potential business opportunities exist. The price categories were used throughout the analysis to evaluate market composition, identify premium markets, and support strategic pricing recommendations.

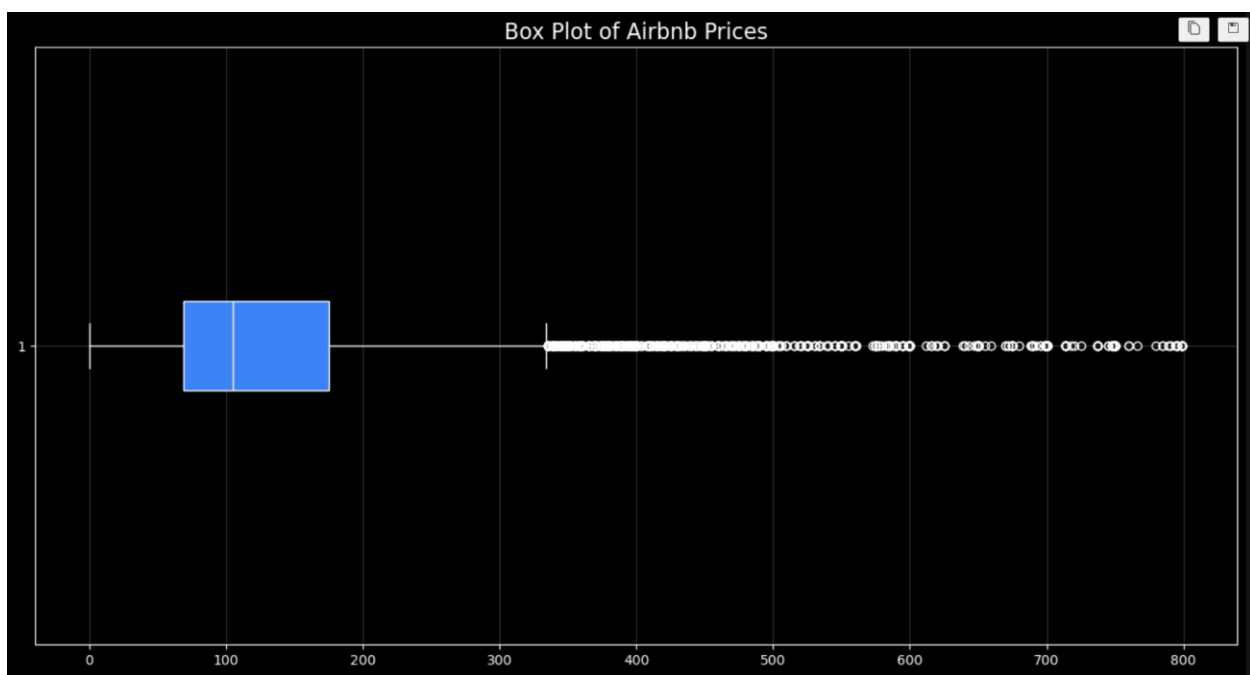
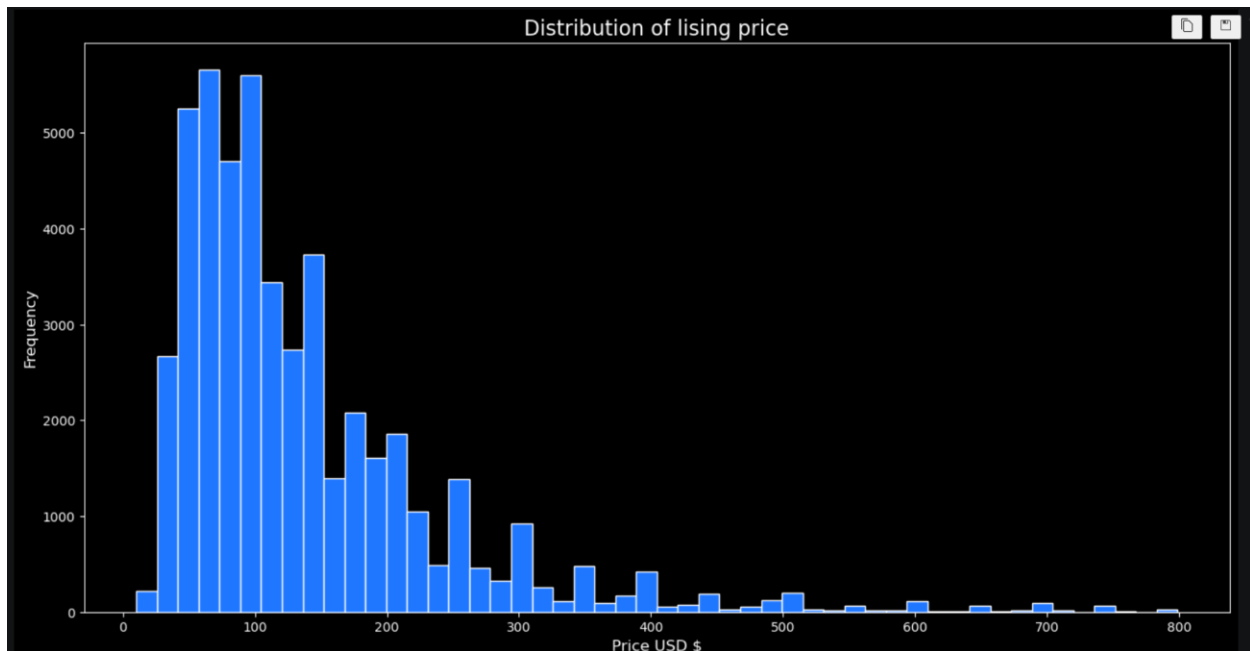
Why reviews_per_month was used as the demand proxy:

The dataset does not contain direct measures of customer demand, such as booking frequency or occupancy rate. Therefore, reviews per month was used as a proxy for demand. Although the dataset includes the total number of reviews, this factor is influenced by how long a listing has been active. Older listings naturally get more reviews over time, making it less suitable for measuring current demand.

In contrast, reviews per month shows the average review activity over time and provides a better understanding of recent customer engagement. While it is not a direct measure of bookings, it serves as a reasonable proxy for comparing demand across listings when occupancy data is unavailable.

5. Exploratory Data Analysis (EDA)

5.1 Price Distribution:

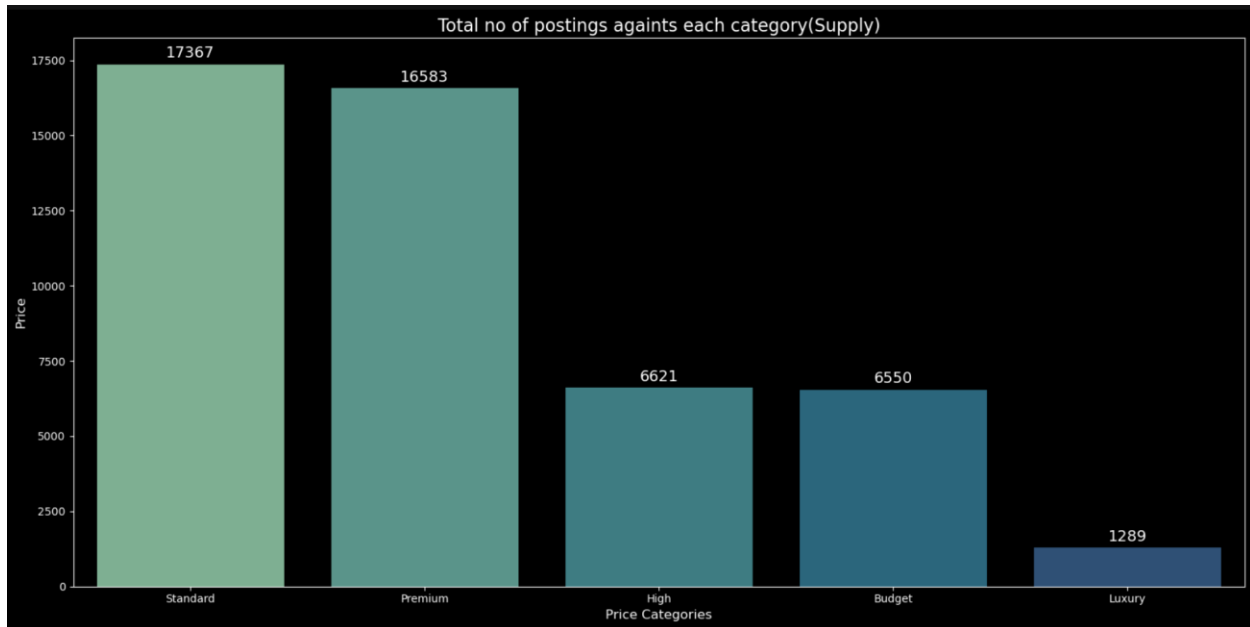


The distribution of the listing prices is right-skewed. Most of the listings falls between 50 and 200\$, indicating a strong concentration in the mid-price range.

Although the extreme price outliers were removed using the **99th percentile**, a small number of higher-priced listings remain. These represent genuine premium properties and are important for understanding the luxury segment of the market.

5.2 Supply Distribution:

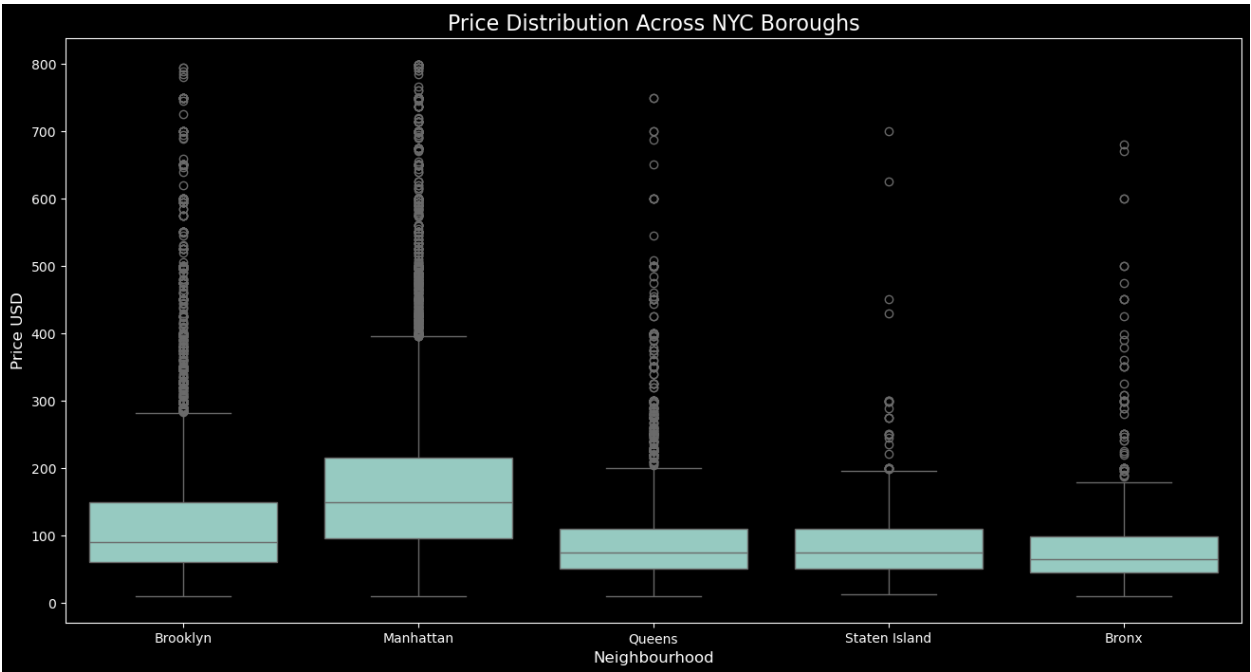
Supply is measured by the total number of listings. This analysis examines which price categories contribute the most to the overall market supply.



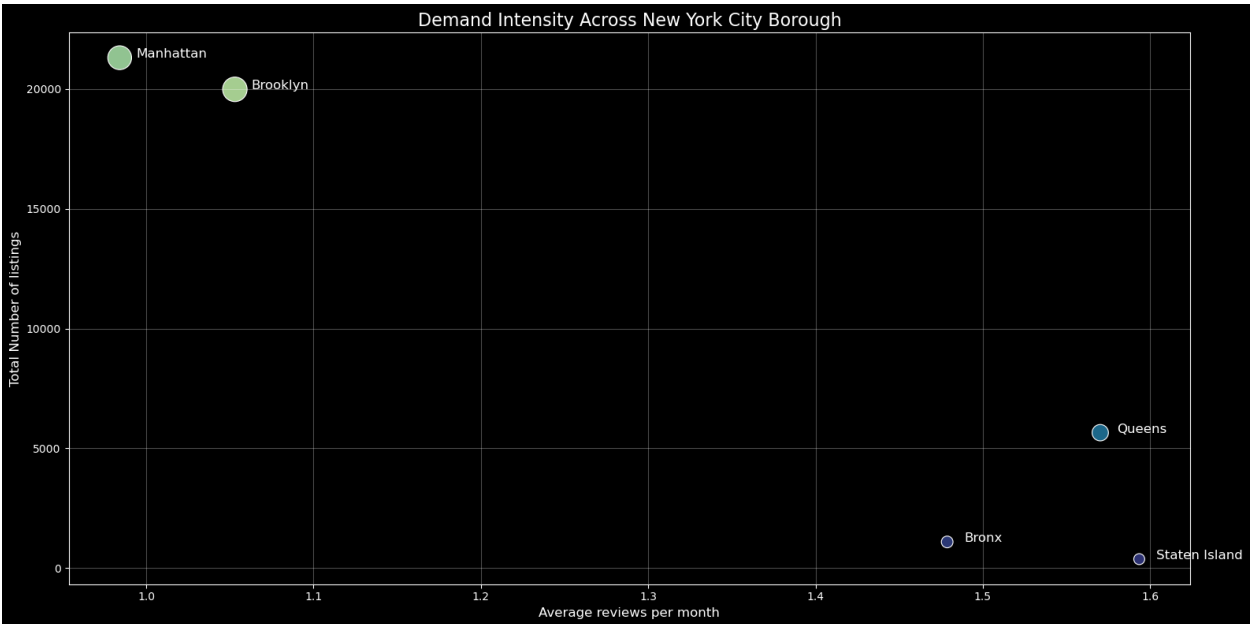
The short term rental market is primarily concentrated within the Standard and Premium price segments which accounts 69% together. Luxury listings represent only a small portion of the marketplace. The concentration of listings within the Standard and Premium segments indicates intense competition in these markets. *Hosts operating in these segments may need to differentiate through location, amenities, and guest experience(reviews).*

5.3 Borough Analysis

Which borough offers the strongest combination of pricing power and demand?



Manhattan exhibits the highest median listing price among all boroughs. While most listings are concentrated between \$100 and \$200 per night, the borough also contains a good number of high-priced listings ranging from \$400 to \$800, indicating a strong premium and luxury market segment. They may have greater flexibility to implement higher pricing strategies and maximize revenue potential

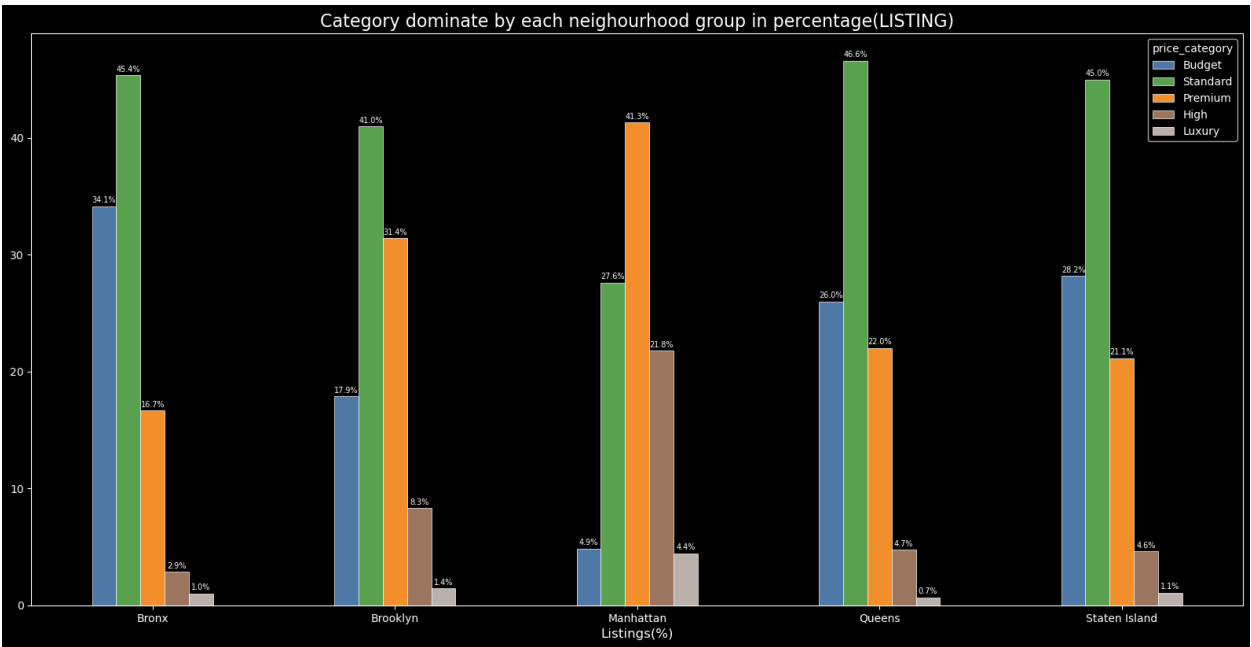


This is the interesting fact here. Based on reviews per month as a proxy for demand, Staten Island, Bronx, and Queens demonstrate the highest review activity per month. Manhattan, despite commanding the highest prices, most number of listings, exhibits lower review activity per month, suggesting that its premium pricing may be associated with lower booking frequency compared to more affordable boroughs.

For the property owners seeking higher booking activity, Staten Island and Bronx may present attractive opportunities due to their strong demand signals and lower price levels compared to Manhattan. While Manhattan remains the premium marketplace because of the higher avg price, it also has highest number of listings, resulting in greater competition among hosts.

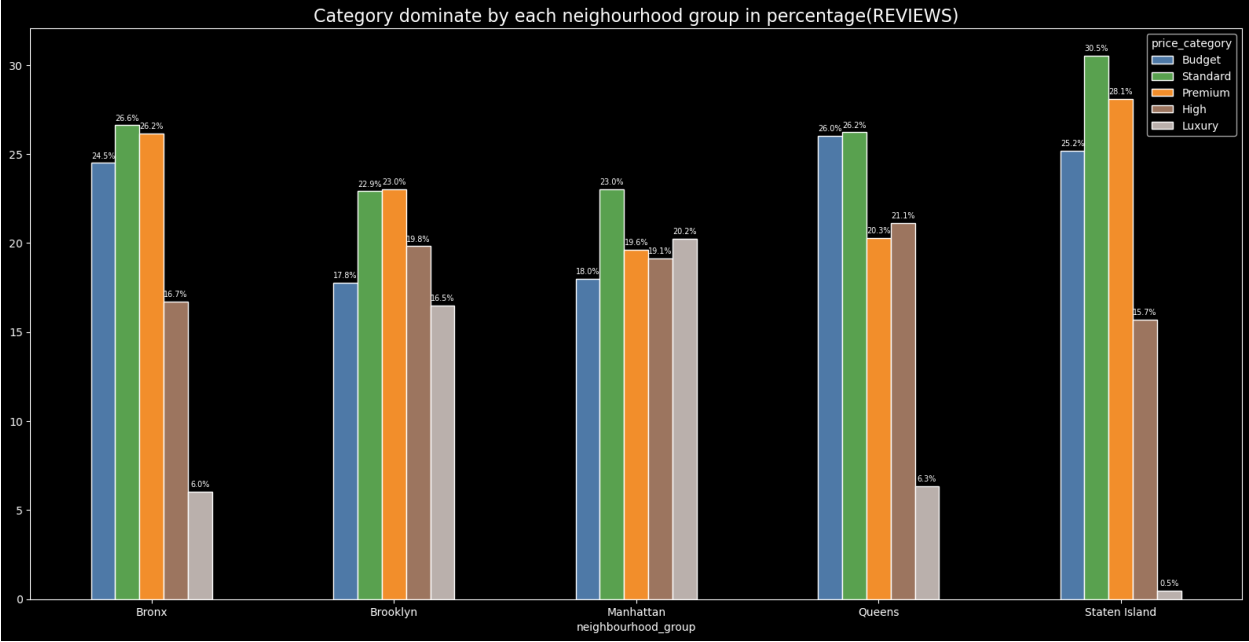
5.4 Market Opportunity:

This section involves the relationship between market supply and customer demand. The objective of this section is to find the premium market, emerging growth opportunities and areas with potential for portfolio expansion.



Most neighbourhoods are dominated by the standard and premium listings, showing the mid range options are the most common. Manhattan has more high prices listings because budget listings goes down and there was an increase in premium, high and luxury listings as compared to other areas. Bronx, queen and Staten Island focus more on affordable options like budget and Standard.

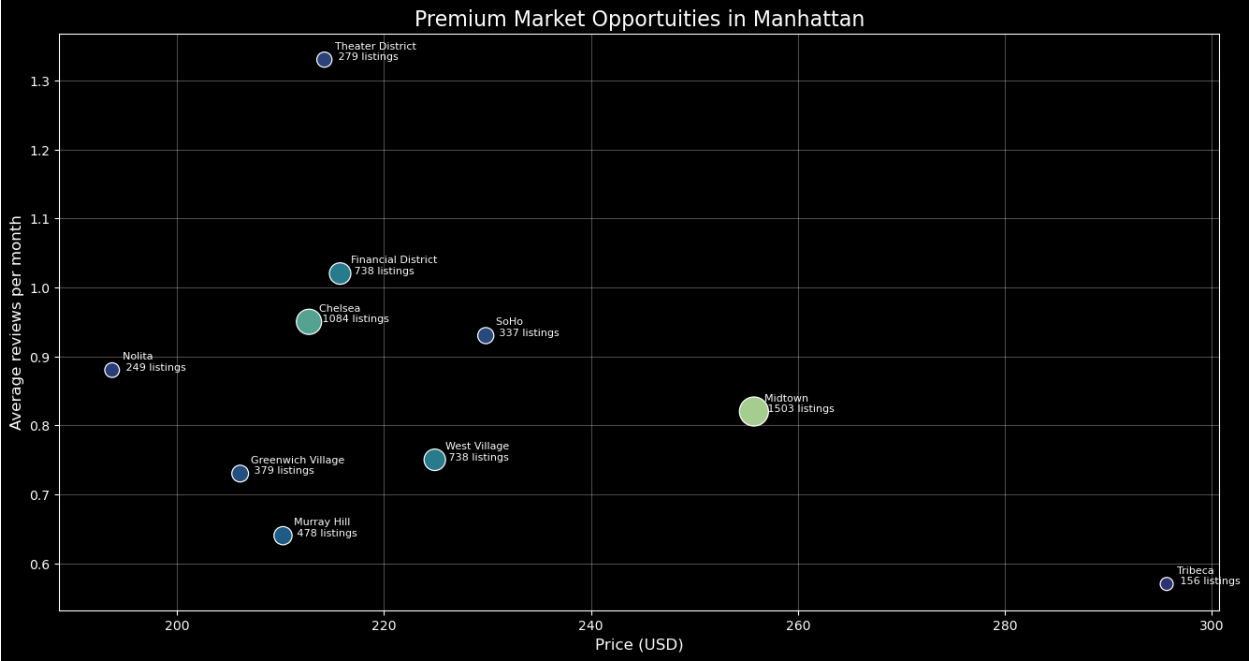
Based on the listings, In Manhattan, Premium and high ending listings are well supported accounted 63%, making it best option for high revenue per booking. However, for Staten, Bronx and Queens, Standard and Budget dominated, indicating stronger opportunities for value focused listings.



Staten Island shows strong demand despite having fewer listings, we can see that Standard and Premium segments contributes to the majority of customer engagement across most neighbourhoods. Manhattan remains the strongest market for higher-priced categories like Luxury listings.

5.5 Neighborhood Drill-down

Manhattan was selected for further analysis because it achieved the highest **Market Opportunity Score**, reflecting the strongest combination of premium pricing, market size, and customer demand. This makes it the most attractive borough for identifying neighbourhood-level investment opportunities and supporting portfolio expansion decisions.



While Tribeca holds the premium pricing with lower review activity and less listings, Financial District, Soho, Chelsea and Midtown commands the balance between pricing, inventory and customer engagement. While neighborhoods such as West Village and Murray Hill maintain moderate pricing, they generate comparatively lower review activity, suggesting lower guest engagement than the leading markets.

Prioritize expansion in Midtown, Chelsea and Financial District, where premium pricing is supported by strong demand and large market size, while treating Tribeca as a luxury niche.

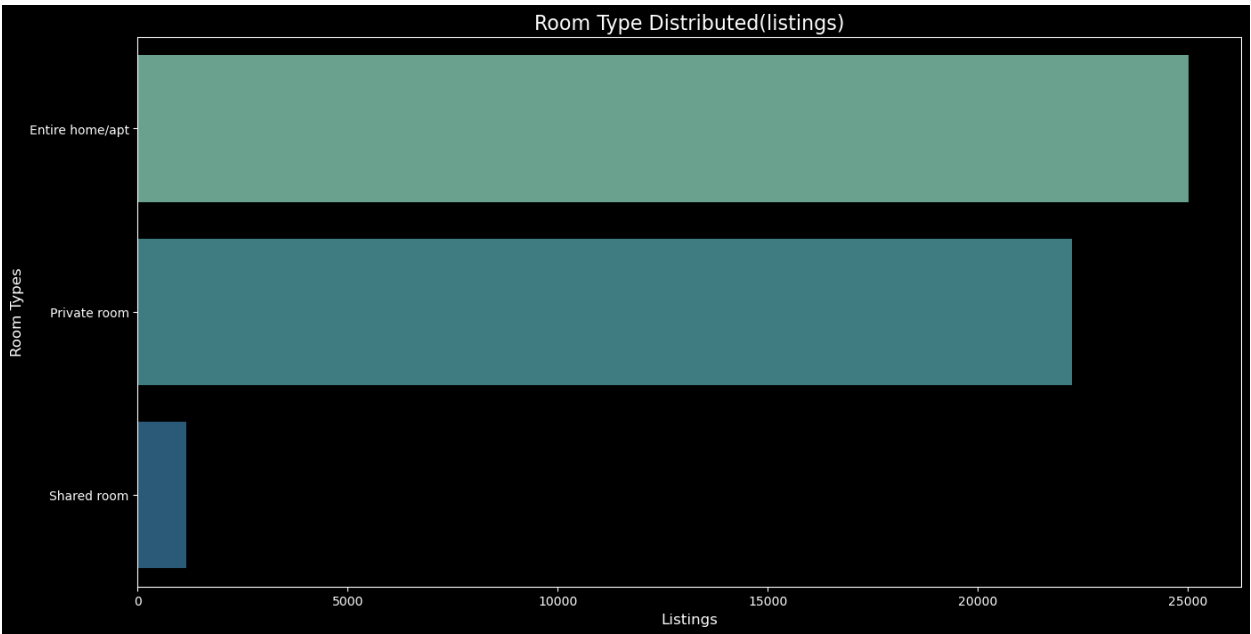
	avg_price	listing_count	avg_reviews	Market Type	recommendation
neighbourhood					
Tribeca	295.65	156	0.57	Luxury Niche	Optimize Pricing
Midtown	255.75	1503	0.82	Luxury Niche	Optimize Pricing
SoHo	229.83	337	0.93	Premium Growth	Expand
West Village	224.90	738	0.75	Stable Market	Maintain
Financial District	215.75	738	1.02	Premium Growth	Expand
Theater District	214.23	279	1.33	Premium Growth	Expand
Chelsea	212.74	1084	0.95	Premium Growth	Expand
Murray Hill	210.23	478	0.64	Stable Market	Maintain
Greenwich Village	206.09	379	0.73	Stable Market	Maintain
Nolita	193.71	249	0.88	Stable Market	Maintain

Based on the table above, The Property Management Team should prioritize portfolio expansion in SoHo, Financial District, Theatre District and Chelsea, where premium pricing and healthy customer demand indicate strong growth potential. For Tribeca and Midtown, focus on pricing optimization rather than expansion, as these neighborhoods already command premium prices but exhibit comparatively lower demand. Stable markets such as West Village, Murray Hill, Greenwich Village, and Nolita should be maintained and monitored for future demand changes.

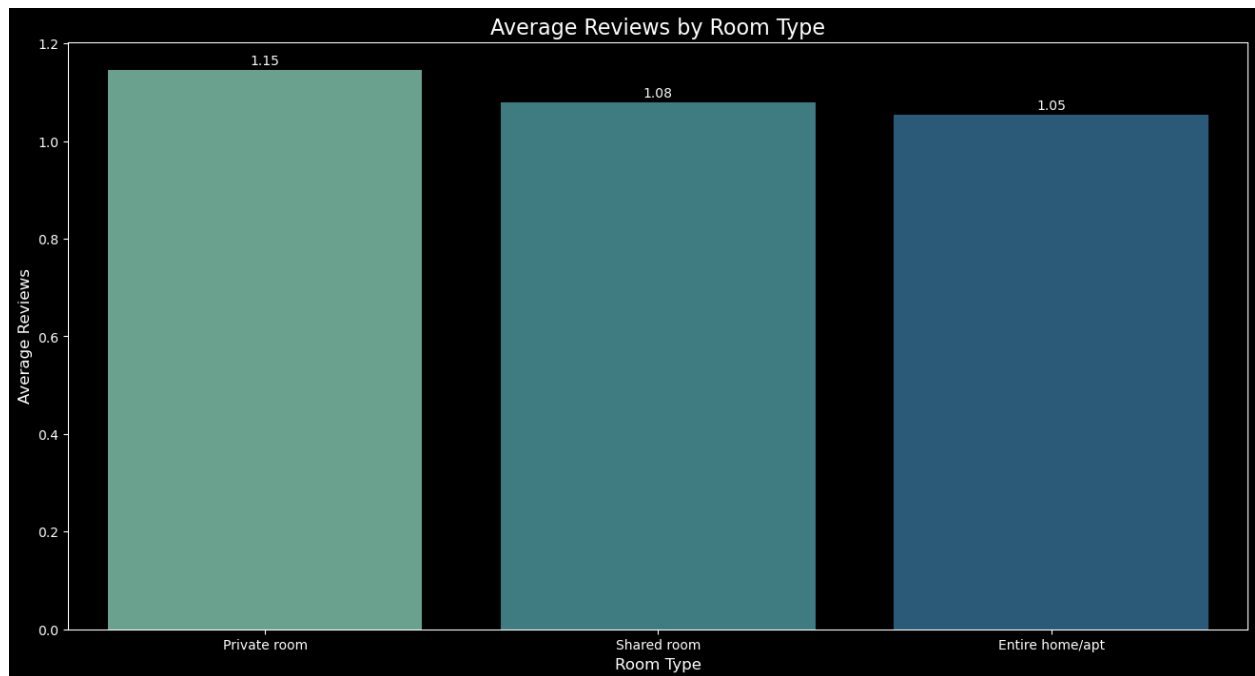
5.6 Room Type Analysis

Having identified the most promising location, the analysis now focus more on the room types. This help determine which accommodation type delivers the best combination of pricing, customer demand and market availability.

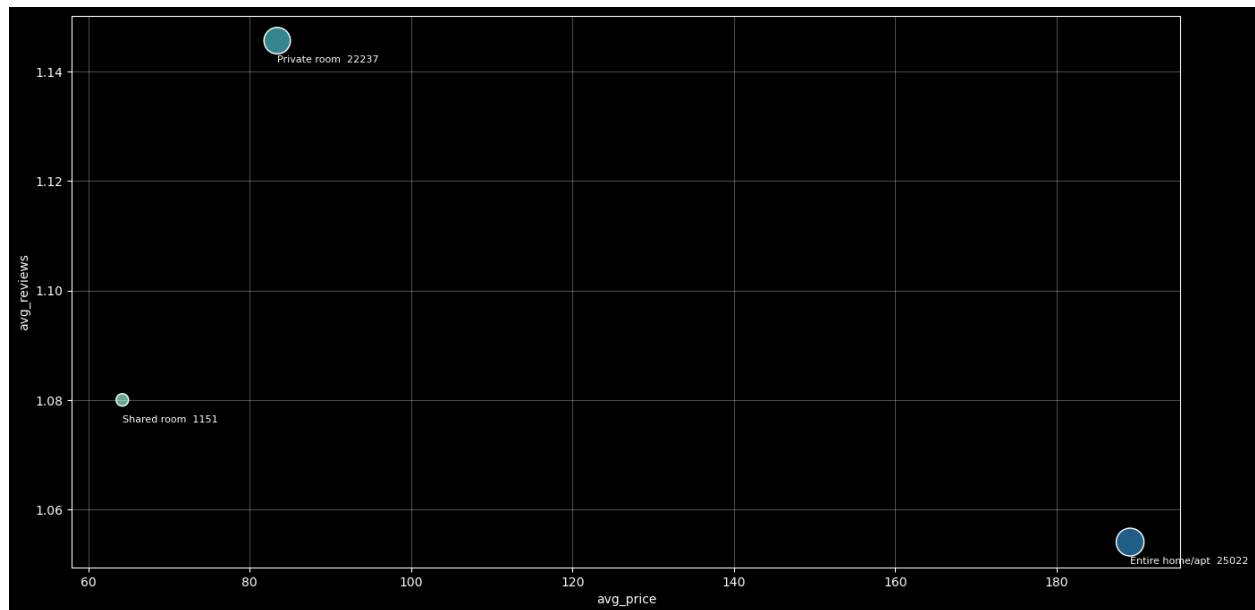
The first step is to compare average prices across different room types to understand which accommodation types generate the highest revenue potential.



This shows that Entire home/apartment listings dominate the Airbnb market, followed by Private rooms, while Shared rooms make up only a small portion of total listings. This suggests that most hosts and customers prefer more private accommodation options.



Despite having lower supply than the Entire home/apt, private room generate the highest review activity, suggesting stronger guest engagement.



Private rooms generate the highest customer engagement despite moderate pricing, suggesting that guests prefer a balance between affordability and privacy. Entire homes command premium prices but show slightly lower review activity, while shared rooms attract the low engagement despite being the most affordable option.

Hosts seeking higher occupancy should prioritize private-room listings, as they appear to offer the strongest balance between demand and pricing. Premium-focused investors may target entire-home listings, while shared rooms may require additional factors beyond price to attract demand

6. Machine Learning

Problem

One of the key challenges for the Property Management Team is setting competitive listing prices. Pricing a property too high may reduce bookings, while pricing it too low can lead to lost revenue. Therefore, a machine learning model was developed to predict Airbnb listing prices based on property characteristics.

Features Used

The model was trained using listing characteristics such as:

- Room Type
 - Minimum Nights
 - Longitude
 - Latitude
 - Reviews per Month
 - Calculated Host Listings Count
 - Availability (365 days)
-

Models Evaluated:

Linear Regression: Linear Regression was evaluated as a baseline model and achieved an R2 score of 0.52 only, indicating it explained only 52% of the variation in the airbnb price. It is suggested that it is influenced by complex non linear relationships between different characteristics.

Random Forest Model: The Random Forest model demonstrated a significant improvement over the baseline Linear Regression model by effectively capturing non-linear relationships within the data. With an R² score of 0.60 and lower prediction errors, it proved to be a strong predictive model for Airbnb price estimation.

XGBOOST Model: This model shows solid generalization with small gap between train, test, and cross-validation scores with MAE: 43.5 RMSE:76.89 R2 Score: 0.63 (Matches the average CV score of 0.63) and was identified as the best model with the help of hyperparameter Tunning. Hyperparameter tuning

improved the XGBOOST model's predictive performance, reducing prediction errors and increasing the R^2 score.

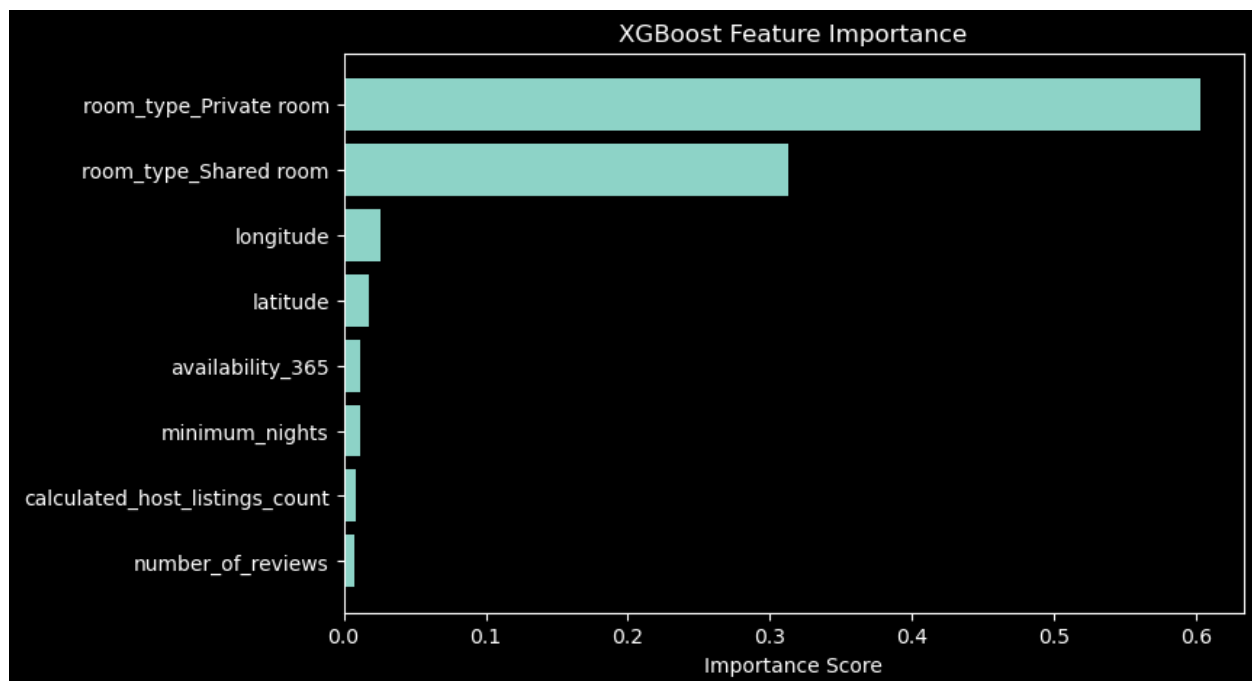
Prediction:

To demonstrate the practical application of the final model, a sample Airbnb listing was created using specific property characteristics, including room type, minimum nights, availability, host listing count, location coordinates, and review count.

After applying the same preprocessing steps used during training, the tuned model predicted an estimated nightly Airbnb price of 77.86.

Feature Importance:

Room type is by far the strongest driver of Airbnb prices, accounting for over 90% of the model's predictive importance. Location factors (latitude and longitude) have a smaller but noticeable influence, while availability, minimum stay requirements, host portfolio size, and review volume contribute relatively little to pricing decisions.



Conclusion:

This project analyzed Airbnb listings across New York City to identify the factors influencing pricing, demand, and investment opportunities. Through exploratory data analysis, Manhattan emerged as the city's primary premium market due to its high pricing and large inventory, while Staten Island showed potential as an emerging growth market because of its strong customer demand despite limited supply. A neighbourhood-level analysis further identified areas such as **Midtown, Chelsea, Financial District, and Theater District** as attractive locations for portfolio expansion based on their balance of pricing, demand, and market size.

The analysis also showed that **Entire home/apartment** listings command the highest prices, while **Private rooms** generate stronger customer engagement. These findings provide valuable insights for pricing strategies and portfolio planning.

To support decision-making, a **Market Opportunity Score** was developed by combining pricing, demand, and market size, helping prioritize boroughs for investment. In addition, machine learning models were built to predict Airbnb listing prices, with **XGBoost** achieving the best performance (**Cross-Validation R^2 = 0.63**, **Test R^2 = 0.65**). Although the model does not capture every factor affecting price, it provides a useful decision-support tool for estimating competitive listing prices.

Overall, this analysis demonstrates how data-driven insights can support property managers in identifying profitable markets, optimizing pricing strategies, and making informed investment decisions.